LectureLens

AI-Powered Lecture Capture System

Complete Documentation & User Guide

Version 1.0
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1. Overview

LectureLens is a plug-and-play AI lecture capture system that runs entirely from a USB drive. It records classroom whiteboards and projector screens using a ceiling-mounted webcam, automatically generates detailed AI notes using Claude Sonnet, answers instructor questions in real time, and produces three professionally formatted HTML files at the end of every session — all without requiring any software installation on the school computer.

1.1 What it does

- Captures photos of the whiteboard at the press of a button or presenter remote
- Takes screenshots of slides, websites, or any screen content with the S key
- Sends each capture to Claude AI using a smart thinking prompt that identifies content type before writing notes
- Smart Lookback automatically compares captures and removes blurry or duplicate shots
- Allows the teacher to type questions (N key) or photo-questions (O key) and receive AI answers
- Saves all notes, photos, and Q&A to a dated session folder on the USB
- Generates three HTML files with full LaTeX math rendering via MathJax when the session ends
- Recovers gracefully if the laptop closes early using backup text files

1.2 Who it is for

LectureLens is designed for teachers and educators who want to provide their students with accurate, detailed lecture notes without the effort of manual note-taking. It is especially useful for subjects involving equations, diagrams, and complex whiteboard content such as mathematics, physics, chemistry, and engineering.

2. Hardware Requirements

2.1 What you need

Item	Purpose	Estimated Cost
Webcam (1080p or higher)	Captures whiteboard from ceiling	Already owned
Active USB extension cable (5m+)	Connects camera from ceiling to computer	~$12
Wireless presenter remote (clicker)	Teacher triggers captures wirelessly	~$20
Drop ceiling camera bracket	Mounts camera to ceiling T-bar grid	~$12
USB flash drive (32GB+)	Runs the program and stores all data	~$10
School computer with internet	Runs the exe and sends API calls	Already available

2.2 Camera recommendation

Any 1080p webcam with a USB connection will work. For best results in a classroom ceiling mount, use a camera with a wide field of view (90–120 degrees) and good low-light performance. Version 2.0 includes CLAHE adaptive brightness correction that automatically adjusts for dark or unevenly lit classrooms, so camera low-light performance is less critical than in Version 1.0.

2.3 Physical setup

1. Mount the camera bracket to the drop ceiling T-bar grid above the whiteboard
2. Attach the webcam to the bracket using the standard 1/4-inch tripod screw
3. Run the USB extension cable along the ceiling and down the wall to the computer
4. Plug the camera extension cable into the computer USB port
5. Plug the presenter remote USB dongle into another USB port
6. Plug the LectureLens USB drive into a third USB port
7. Double-click LectureLens.exe on the USB drive to start

The camera stays mounted permanently. Only the USB drive and presenter remote travel with the teacher between sessions.

3. Software & Files

3.1 What is on the USB drive

File	Description
LectureLens.exe	Main program — double-click to start a lecture session
Recover.exe	Recovery tool — rebuilds HTML files from backup notes if laptop closed early
sessions/	Folder where all session data is automatically saved

3.2 Session folder structure

Every lecture creates a new dated and named folder inside sessions/. Claude names it based on the lecture content after the session ends:

Path	Contents
sessions/2026-05-16_09-30_linear-algebra-matrices/	Root session folder (renamed by Claude after session)
01_photos.html	All captured whiteboard photos only (no screenshots)
02_ai_notes.html	Claude's notes for every capture + screenshots + Q&A in chronological order
03_summary.html	Full lecture summary with LaTeX math rendered by MathJax
images/	Raw JPEG/PNG files for every photo and screenshot captured
backups/	Text files saved after each Claude response (crash recovery)
backups/replaced_captures/	Captures removed by Smart Lookback (kept as backup)

3.3 The three HTML files explained

01_photos.html

Contains every raw whiteboard photo captured during the lecture, one per section, with a timestamp header. Screenshots taken with the S key are NOT included here — they appear in 02_ai_notes.html only. Open in any browser to review exactly what was on the board at each point.

02_ai_notes.html

Contains Claude's detailed notes for every capture in chronological order — exactly as the lecture progressed. Camera captures (blue header), screenshots (green header), text notes N key (orange header), and photo notes O key (orange header) all appear in the order they were taken. All math is rendered as proper typeset equations via MathJax. Open in any browser — print to PDF using Ctrl+P and Save as PDF for a printable version.

03_summary.html

Contains a comprehensive lecture summary generated by Claude after all captures are processed. Includes the main topic, learning objectives, key equations, step-by-step procedures, major takeaways, and a Questions and Answers from Class section. All math rendered with MathJax. Open in any browser.

4. Controls & Keyboard Shortcuts

4.1 Key reference

Key / Button	Action	Details
SPACE	Capture photo	Takes a photo instantly, queues for Claude processing
Remote clicker (right arrow)	Capture photo	Same as SPACE — works from anywhere in the room
iClicker button (F5)	Capture photo	Multiple iClickers can be used simultaneously
S	Screenshot + note	Captures full screen, preview minimizes, optional note typed, Claude analyzes content type and writes notes
N	Text note	Preview minimizes, type a question, Claude answers in terminal and HTML
O	Photo note	Snaps board photo, preview minimizes, type question, Claude answers with board context
TAB	Switch camera	Toggles between ceiling cam and doc cam (if second camera detected)
ESC	Stop session	Waits for queue to finish, generates all 3 HTML files automatically

4.2 Capture workflow (SPACE / remote)

8. Photo taken instantly and saved to images/ folder with CLAHE brightness correction
9. Green flash border appears on the live preview window
10. Photo added to the processing queue
11. Claude processes it in the background using the 4-step thinking prompt (3–8 seconds)
12. Smart Lookback compares the new capture against recent ones — removes blurry or duplicate captures
13. Notes saved to backups/ folder as a text file
14. Live preview shows updated counts: Done: X Queue: Y

4.3 Screenshot workflow (S key) — NEW in v2.0

15. Full screen captured silently and saved to images/ folder
16. Preview minimizes automatically
17. Terminal asks: Add a note about this screenshot? (press Enter to skip — auto-cancels in 15s)
18. Claude identifies what type of content is shown (slide, website, code, document, etc.)

19. Claude writes structured notes based on the content type
20. Preview restores automatically
21. Screenshot and notes appear in 02_ai_notes.html with a green header

4.4 Note workflow (N key)

22. Preview minimizes automatically
23. Terminal asks for your note or question
24. Type your question and press Enter — auto-cancels in 15 seconds if no real characters typed
25. Spaces alone do not reset the timer — only real characters count
26. Claude generates a written answer (~3 seconds)
27. Answer appears in the terminal
28. Note and answer saved to backups/ and included in 02_ai_notes.html
29. Preview restores automatically

4.5 Photo note workflow (O key)

30. Board photo taken silently
31. Preview minimizes automatically
32. Type your question and press Enter
33. Claude answers using both your question AND the board photo for full context
34. Answer appears in terminal with board context label
35. Preview restores automatically

4.6 Smart Lookback — NEW in v2.0

After every capture, Smart Lookback automatically compares the newest photo against the last 3 captures. If an older capture is blurry, unclear, or a duplicate of the newer one, it is hidden from the final HTML files. The hidden capture is moved to backups/replaced_captures/ and kept as a backup — it is never permanently deleted.

- Runs automatically after every capture — no user action needed
- Also uses mathematical reasoning to fill in [unclear] values where possible
- Can be disabled by setting SMART_LOOKBACK_REPAIR = False in the script
- Adds approximately $0.05 per capture to the API cost

5. Cost Breakdown

5.1 Hardware — one-time cost

Item	Cost
Active USB extension cable (5m)	~$12
Wireless presenter remote	~$20
Drop ceiling camera bracket	~$12
USB flash drive 32GB	~$10
Total one-time hardware	~$54

5.2 API cost per lecture (Version 2.0)

The following costs are based on a typical lecture with 30 photos, 4 notes (N/O keys), and 2 screenshots (S key). Claude Sonnet pricing: $3.00 per million input tokens, $15.00 per million output tokens.

API Call	Model	Tokens (in/out)	Cost
Session naming	Haiku	~70 / 20	~$0.00
30 photo captures — input (images)	Sonnet	30 × 1,500 = 45,000	$0.13500
30 photo captures — output (notes)	Sonnet	30 × 1,200 = 36,000	$0.54000
Smart Lookback (after each capture)	Sonnet	30 × 4,500 = 135,000 in / 30 × 800 out	$0.40500 + $0.36000
4 text notes N key	Sonnet	4 × 800 in / 4 × 500 out	$0.00960 + $0.03000
4 photo notes O key	Sonnet	4 × 1,800 in / 4 × 600 out	$0.02160 + $0.03600
2 screenshots S key	Sonnet	2 × 3,000 in / 2 × 1,600 out	$0.01800 + $0.04800
Final HTML summary	Sonnet	~40,000 in / 3,000 out	$0.12000 + $0.04500
Folder rename	Haiku	~70 / 20	~$0.00
HTML generation (3 files)	—	No API call	FREE
Total per lecture			~$1.70

5.3 What $20 of API credit gets you

Photos per Lecture	Notes + Screenshots	Cost per Lecture	Lectures from $20	Weeks (2x/week)
30 photos	4 notes, 2 screenshots	~$1.70	~11 lectures	~5.5 weeks
30 photos	0 notes, 0 screenshots	~$1.10	~18 lectures	~9 weeks
30 photos (Smart Lookback OFF)	4 notes, 2 screenshots	~$0.91	~22 lectures	~11 weeks
50 photos	4 notes, 2 screenshots	~$2.60	~7 lectures	~3.5 weeks

Tip: Disable Smart Lookback (SMART_LOOKBACK_REPAIR = False) to reduce cost by ~60% while still getting full notes. Re-enable it for important lectures where quality matters most.

6. Recovery — If the Laptop Closes Early

6.1 How data is protected

LectureLens saves backup notes to the USB drive after every single Claude response. If the laptop is closed, the power goes out, or the program crashes, all completed notes are already on the USB as text files in the backups/ folder. No internet connection is required to recover them.

Data type	Saved?	When
Raw photos (images/)	Yes — always	Instantly when SPACE is pressed
Screenshots (images/)	Yes — always	Instantly when S is pressed
AI notes (backups/)	Yes — always	As soon as Claude responds (~5 sec)
Instructor notes (backups/)	Yes — always	As soon as note is saved
Smart Lookback results (backups/)	Yes — always	After each lookback check
3 HTML files	Only if ESC was pressed	Generated at end of session
Photos still in queue	No	Lost if laptop closes mid-processing

6.2 Using Recover.exe

36. Plug the USB into any Windows computer
37. Double-click Recover.exe
38. A list of all sessions appears — pick the one to recover
39. Recover.exe reads all backup text files and builds 3 HTML files automatically
40. No internet connection or API key required
41. HTML files appear in the session folder ready to open in any browser

Note: The summary HTML created by Recover.exe is a compiled version of all backup notes rather than a new AI-generated summary. The content is identical — just compiled without a new Claude call.

7. Multiple Cameras

7.1 How camera detection works

LectureLens automatically scans for all connected USB cameras when it starts. If more than one is detected, they are listed in the terminal with labels. The first camera is labeled ceiling/main and the second is labeled doc cam.

7.2 Switching cameras

Press TAB at any time during the session to toggle between cameras. The live preview shows the active camera name in the corner. Captures always use whichever camera is currently active.

7.3 Using with a document camera

If the teacher has a doc cam connected (for example through a Steam Deck or direct USB), it appears as Camera 1 automatically. The teacher can switch to it mid-lecture to capture documents or textbook pages, then switch back to the ceiling camera.

8. Multiple Clickers & Student Participation

8.1 How multiple remotes work

Any number of USB presenter remotes or iClicker devices can be plugged in simultaneously. Windows merges all USB keyboard input so pynput sees every press regardless of which device sent it. The 3-second cooldown prevents duplicate captures if multiple people press at the same time.

8.2 Supported devices

- Logitech wireless presenter remotes — right arrow button triggers capture
- iClicker remotes — F5 button triggers capture
- Any USB keyboard remote — spacebar triggers capture

8.3 Multi-clicker classroom setup

42. Teacher holds one clicker — captures key moments throughout the lecture
43. 1–2 students hold additional clickers — capture moments they want noted
44. 3-second cooldown prevents spam if two people press simultaneously
45. All captures from all clickers are identical and appear in the same HTML files

9. Troubleshooting

Problem	Likely Cause	Solution
Exe does not open	Program still running in background	Close the terminal window first, then try again
No camera detected	Camera not plugged in or wrong USB port	Unplug and replug the camera, restart the exe
No HTML files generated	Session ended before Claude finished	Use Recover.exe to rebuild all 3 HTML files from backups
Smart Lookback removing too many captures	Threshold too aggressive	Set SMART_DELETE_WORSE_CAPTURES = False to keep all captures
N/O/S key does nothing	Preview window focused instead of terminal	Click the terminal window once, then press the key
Note auto-cancelled immediately	Only spaces typed — spaces do not reset timer	Type at least one real letter or number to activate the timer
Preview frozen showing INPUT MODE	Note mode active	Type your note and press Enter, or wait 15 seconds to auto-cancel
Math not rendering in HTML	JavaScript disabled in browser	Enable JavaScript in browser settings or try a different browser
Preview window is black	Camera in use by another app	Close AverMedia or other camera apps, restart exe

10. Quick Start Guide for Teachers

This page can be printed and kept with the USB drive for reference.

Starting a session

- 1. Plug in the USB drive, camera, and presenter remote
- 2. Double-click LectureLens.exe on the USB drive
- 3. Wait for Camera ready. Waiting for input... to appear
- 4. Check the live preview window to confirm the board is visible

During the lecture

- SPACE or remote clicker = capture what is on the board
- S key = screenshot the full screen (slides, websites, documents)
- N key = type a text question — Claude answers without seeing the board
- O key = type a question — Claude sees the board and answers with context
- TAB = switch between ceiling camera and document camera
- Preview shows: Done: X Queue: Y so you always know status

Ending a session

- 1. Press ESC — the program waits for all captures to finish processing
- 2. Wait for All captures processed! and the HTML generation messages
- 3. Press Enter when prompted — the program closes
- 4. Open the sessions/ folder on the USB to find your 3 HTML files
- 5. Open any .html file in Chrome or Edge for full math rendering
- 6. To print: Ctrl+P then Save as PDF in the browser
- 7. Unplug the USB drive safely

If the laptop closes early: plug USB into any computer and double-click Recover.exe to rebuild all 3 HTML files from saved backup notes. No internet required.